



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
08-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

PAPER-1

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

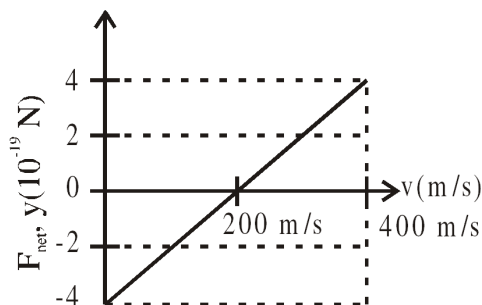
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. At time ' t_0 ' an electron is sent along the positive direction of x axis, through both an electric field $\vec{E}$ and a magnetic field $\vec{B}$, with $\vec{E}$ directed parallel to the 'y' axis, graph gives the y component ($F_{\text{net}, y}$) of the net force on the electron due to two fields as a function of the electron speed ' v ' at time t_0 . Assuming $B_x = 0$. Find magnitude of electric field $\vec{E}$ in (N/C). Use ($e = 1.6 \times 10^{-19}$ C) (The x & z components of net force are zero at time t_0)



(A) 1.25 N/C

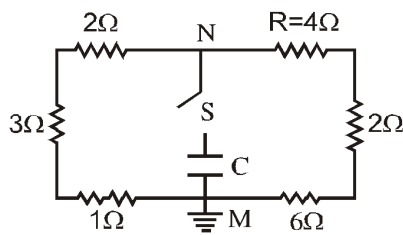
(B) 5 N/C

(C) 2.5 N/C

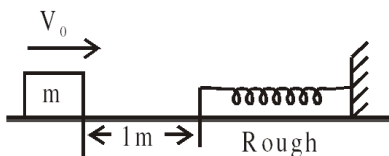
(D) 10 N/C

Space for Rough Work

2. A capacitor of capacity $8\mu\text{F}$ and the initial charge $160\mu\text{C}$, is connected with a key S , and different resistors as shown in figure. Point M is earthed. If key is closed at $t = 0$ then the current through resistance ($R = 4\Omega$) at $t = 32\mu\text{s}$, is



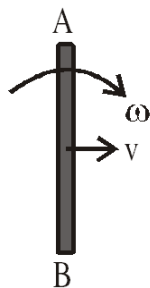
- (A) $10/3e$ (B) $5/3e$ (C) $5/e$ (D) $10/e$
3. A block of mass $m = 2\text{kg}$ is moving with velocity v_0 towards a massless, unstretch spring of force constant $K = 10\text{ N/m}$ on a rough surface as shown. Coefficient of friction between the block and the ground is $\mu = 1/2$. Find the maximum value of v_0 so that after compressing the spring, the block does not return back but stops there permanently.



- (A) $\sqrt{6.4}\text{ m/s}$ (B) 5 m/s (C) $\sqrt{20}\text{ m/s}$ (D) $\sqrt{5}\text{ m/s}$

Space for Rough Work

4. A metal rod of length ℓ , moving with an angular velocity ω and velocity of its centre is v . Find potential difference between points A and B at the instant shown in figure. A uniform magnetic field of strength B exist perpendicular to plane of paper:



- (A) $Bv\ell$ (B) $Bv\ell + \frac{1}{2}B\omega\ell^2$
 (C) $B\omega\ell - \frac{1}{2}B\omega\ell^2$ (D) $Bv\ell + B\omega\left(\frac{\ell}{2}\right)^2$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

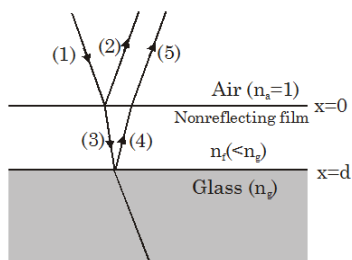
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. If a film (having suitable thickness d) with a refractive index less than that of glass is coated on the glass, then waves reflected from the upper surface of the film interferes destructively with the waves reflected from the lower surface of the film. Such a film is called non-reflecting film. Let n_a , n_f and n_g be the refractive index of air, reflecting film, and glass, respectively. Select the correct alternative. (λ is the wavelength of light in air)



(A) $d = \frac{\lambda}{4n_f}$

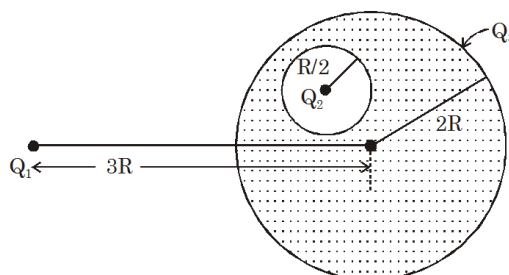
(B) $d = \frac{\lambda}{2(2n_f - 1)}$

(C) For complete destructive interference, $n_f \approx \sqrt{n_a n_g}$, (when $n_a \approx n_g$)

(D) For complete destructive interference, $n_f \approx \frac{n_a + n_g}{2}$, (when $n_a \approx n_g$)

Space for Rough Work

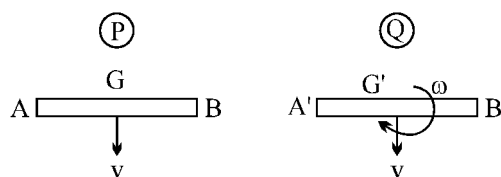
6. A spherical conductor of radius $2R$ has a spherical cavity of radius $\frac{R}{2}$. The cavity does not enclose the centre of sphere. Charges Q_1 and Q_2 are placed as shown in figure. Q_2 is at the center of cavity. An additional charge Q_3 is given to the sphere.



- (A) The potential of the sphere is $\frac{KQ_1}{3R} + \frac{K(Q_2 + Q_3)}{2R}$
- (B) The potential inside the cavity at a distance r ($r < \frac{R}{2}$) from the center of cavity is $\frac{KQ_2}{r} + \frac{5KQ_1}{6R} - \frac{3KQ_2}{2R}$
- (C) The value of potential outside the sphere at a distance r from the center of sphere is $\frac{k(Q_2 + Q_3)}{r} + \frac{kQ_1}{r'}$ where r' is the distance from Q_1
- (D) The charge that will flow into ground if the sphere is grounded is $Q_2 + Q_3 + \frac{2Q_1}{3}$

Space for Rough Work

7. Two equal uniform rods P and Q move with the same velocity v as shown in the figure. The second rod has an angular velocity ω ($< 6v/\ell$) (clockwise) about G' in addition to v .



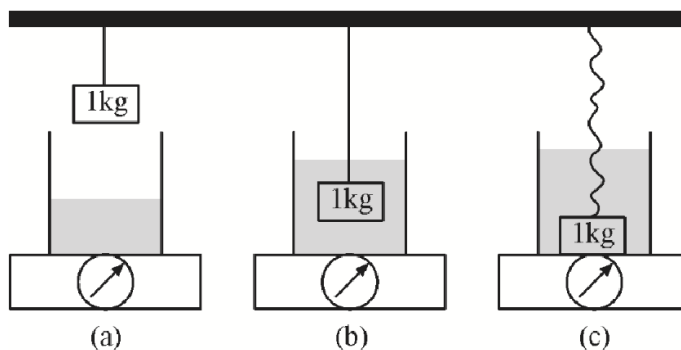
- (A) If the ends A and A' are suddenly fixed simultaneously, both rods will rotate with the same angular velocity
- (B) If the ends A and A' are suddenly fixed simultaneously, the rod Q will rotate with greater angular velocity
- (C) If the ends B and B' are suddenly fixed simultaneously, both rods will rotate with the same angular velocity
- (D) If the ends B and B' are suddenly fixed simultaneously, the rod P will rotate with greater angular velocity
8. Two sources of current of equal e.m.f. are connected in series and have different internal resistances r_1 and r_2 . An external resistance R is connected in series with these sources of e.m.f.
- (A) For $r_1 > r_2$, when $R = (r_1 - r_2)$, the power output of first source is zero.
- (B) When $R = \sqrt{r_1 r_2}$, terminal potential difference to be zero across both sources
- (C) For $r_2 > r_1$, when $R = (r_2 - r_1)$, the power output of second source is zero.
- (D) For $r_1 > r_2$, when $R = (3r_1 - r_2)$, the power output of first source is maximum.

Space for Rough Work

9. The number of divisions on the circular scale of a screw gauge is 50. It moves 0.5 mm on main scale in one revolution. When there is no object between the jaws of instrument and they touch each other, the 5th mark on circular scale coincides with reference line, with zero of circular scale below the reference line. When a ball is placed between the jaws, main scale reading is 4 and 25th mark on circular scale matches with reference line. Select the INCORRECT alternative(s).
- (A) The zero correction is 0.05 mm.
- (B) The diameter of ball must be reported as 2.2 mm
- (C) The least count of the screw gauge is 0.05 mm.
- (D) The diameter of the ball is 2.30 mm
10. Hydrogen atoms in states of high quantum number have been created in the laboratory and observed in space. They are called Rydberg atoms.
- (A) The quantum number of the bohr orbit in a hydrogen atom whose radius is 0.0100 mm is 435
- (B) The quantum number of the bohr orbit in a hydrogen atom whose radius is 0.0100 mm is 400
- (C) The energy of electron in hydrogen atom in this state is -7.19×10^{-5} eV
- (D) The energy of electron in hydrogen atom in this state is 7.19×10^{-5} eV

Space for Rough Work

11. Regarding a satellite revolving around the earth with a time period of 18 hours, in the equatorial plane, select the correct alternative(s)
- (A) The time period of the orbital motion does not depend on the mass of the satellite
- (B) If the satellite revolves in the same sense as that of the rotation of earth, the time interval between its consecutive appearances overhead a fixed point is 72 hrs
- (C) If the satellite revolves in the opposite sense as that of the rotation of earth, the time interval between its consecutive appearances overhead a fixed point is nearly 1.3 hrs.
- (D) If the satellite revolves in the same sense as that of the rotation of earth, the time interval between its consecutive appearances overhead a fixed point is infinite
12. Three identical blocks each of mass $m = 1 \text{ kg}$ and volume $3 \times 10^{-4} \text{ m}^3$ are suspended by massless strings from a support as shown. Underneath are three identical containers containing the same amount of water. That are placed over the scales. In Fig.(a), the block is completely out of the water, in figure (b) the block is completely submerged but not touching the beaker and in figure (c), the block rests on the bottom of the beaker. The scale in figure (a) reads 14 N.



- (A) The tension in the string in (b) is 10 N
- (B) The tension in the string in (b) is 7 N
- (C) The reading of the scale in (b) is 17 N
- (D) The reading of the scale in (c) is 24 N

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

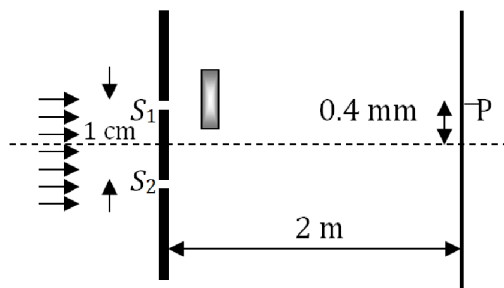
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. A box contains an inductor of inductance L , a capacitor of capacitance C and a resistor of resistance R . When 250 V d. c. is applied to the terminals of the box, a current of 1.0 A plus an oscillating current flows in the circuit. When an a. c. source of 250 V (r.m.s) at 2250 rad/sec is connected, an rms current of 1.25 A is produced. This current rises with frequency and becomes maximum at 4500 rad/sec. Evaluate $2R\sqrt{C/L}$.
 2. An ideal gas containing n moles whose ratio of specific heats is $59/39$ undergoes a change according to the relation $P = P_0 + \alpha/V$ where P_0 and α are positive constants. When the temperature is doubled from an initial value of $2\alpha/nR$, the heat absorbed by the gas is Q , find Q/α . You can use $\ln 3 = 1.10$.
-

Space for Rough Work

3. In a Young's double slit experiment, a plane monochromatic wave of wavelength 600 nm and intensity 10 W/m^2 is incident normally on the slit plane. The length of each slit is 2 mm but their widths are 0.01 mm and 0.04 mm, respectively. A perfectly transparent film of thickness t , and refractive index 1.5 is placed in front of the slit S_1 . All the power of each slit (if it is operational alone) is spread nearly uniformly in a region of breadth 2 mm and height 20 mm above and 20 mm below the central line. The minimum thickness of the film leading to minimum intensity I at the point P is T . Find IT in nano W/m.



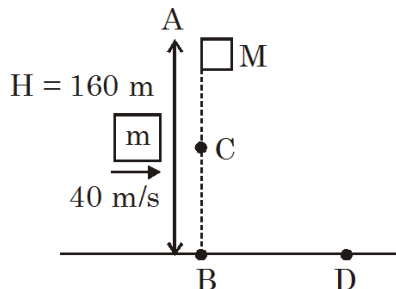
4. A homogenous chain of linear density 0.02 kg/m hangs from two poles located at $x = \pm a$. The equation of the curve formed by the chain is

$$y = b \left(\frac{e^{x/b} + e^{-x/b}}{2} \right)$$

Find its potential energy (in Joules) with respect to the ground, i.e. $y = 0$. Here $b = 2$ meters and $a = b \ln 2$. You can also use the approximation $\ln 2 = 11/16$ and take $g = 10 \text{ m/s}^2$.

Space for Rough Work

5. A body of mass 150 gm is dropped from point A, takes time t_1 seconds to reach the point B at the ground. Once more, the same body is dropped from point A and when body traverses half the distance, a bullet of mass 50gm, moving horizontally with speed 40 m/s hits the falling body at point C and sticks to it. The total time of flight of the body in this case is t_2 seconds and body strikes the ground at point D. The air drag should be neglected (Take $g = 10 \text{ m/s}^2$). Then find the ratio of t_1/t_2 .



6. Two coherent radio-frequency point sources separated by 2.0 m are radiating in phase with $\lambda = 1.0\text{m}$. A detector move in a circular path of large radius around the two sources in a plane containing them. The number of maximum detected by the detector is.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

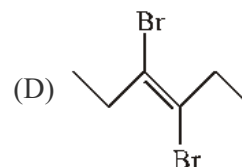
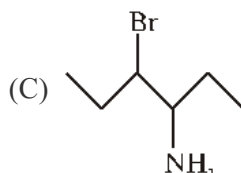
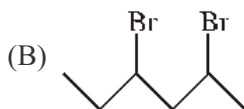
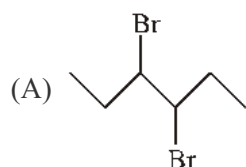
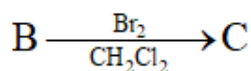
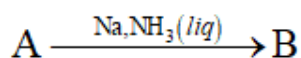
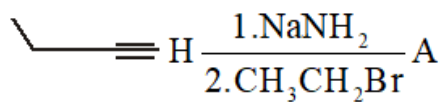
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. In the structure of $B_4O_5(OH)_4^{2-}$:
- (A) All four B atoms are trigonal planar
- (B) One B atom is tetrahedral and the other three are trigonal planar
- (C) Three B atoms are tetrahedral and one is trigonal planar
- (D) Two B atoms are tetrahedral and the other two are trigonal planar
-

Space for Rough Work

2. What is the final product, C, of the following reaction sequence ?



3. The angular momentum of an electron in a Bohr's orbit of H-atom is $3.1652 \times 10^{-34} \text{ kg-m}^2/\text{sec}$. Calculate the wavenumber in terms of Rydberg constant (R) of the spectral line emitted when an electron falls from this level to the ground state. [Use $h = 6.6 \times 10^{-34} \text{ Js}$]

(A) $R \left(\frac{8}{9} \right)$

(B) $R \left(\frac{5}{9} \right)$

(C) $R \left(\frac{7}{9} \right)$

(D) $R \left(\frac{3}{4} \right)$

Space for Rough Work

4. A mixture of chlorides of copper, cadmium, chromium, iron and aluminium was dissolved in water acidified with HCl and hydrogen sulphide gas was passed for sufficient time. It was filtered, boiled and a few drops of nitric acid were added while boiling. To this solution ammonium chloride and sodium hydroxide are added in excess and filtered. The filtrate finally obtained -
- (A) gives test of Cu^{+2} , Fe^{+3}
 - (B) does not gives test of Cd^{+2} , Na^{+} , Cr^{+3}
 - (C) is green coloured but colour change on addition of H_2O_2
 - (D) gives dark green colouration with excess NH_4OH
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Select the correct statement(s) :-

- (A) PhCHO and $\text{CH}_3\text{COCH}_2\text{CH}_3$ gives $\text{PhCH=CHCOCH}_2\text{CH}_3$ in basic medium with heat
- (B) PhCHO and $(\text{CH}_3\text{CO})_2\text{O}$ gives cinnamic acid in aqueous solution of sodium acetate with heat
- (C) 5 chloropentan-2 one in aq NaOH gives cyclopentanone as well as cyclopropylmethyl ketone in unequal quantities with heat
- (D) Only (1) and (2) are correct

Space for Rough Work

6. An ideal gas undergoes through following cyclic process.

1-2 : Reversible adiabatic compression from $P_1V_1T_1$ to $P_2V_2T_2$

2-3 : Reversible isochoric heating from $P_2V_2T_2$ to $P_3V_3T_3$

3-4 : Reversible adiabatic expansion from $P_3V_3T_3$ to $P_4V_4T_4$

4-1 : Reversible isochoric cooling from $P_4V_4T_4$ to $P_1V_1T_1$

Efficiency of the cycle is :

(A) $1 - \left(\frac{T_4 - T_1}{T_3 - T_2} \right)$ (B) $1 - \frac{T_1}{T_2}$ (C) $1 - \left(\frac{V_2}{V_1} \right)^{\gamma-1}$ (D) $1 - \frac{|Q_{4-1}|}{|Q_{2-3}|}$

7. Which of the following ores contains more than one metals ?

(A) Copper pyrite (B) Dolomite (C) Carnalite (D) Cryolite

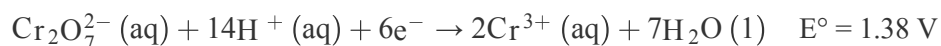
8. Which of the following aryl amines will not form a diazonium salt on reaction with sodium nitrite in hydrochloric acid ?

(A) m-Ethyl aniline (B) p-Amino acetophenone
(C) 4-Chloro-N-Methyl aniline (D) N-Ethyl-2-Methyl aniline

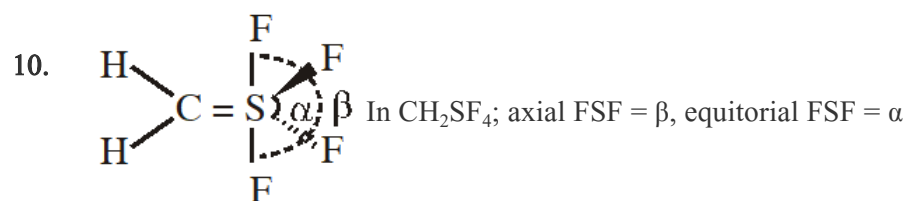
Space for Rough Work

9. Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox reaction.

Some half - cell reactions and their standard potential are given below :



Identify the correct statement(s) regarding the quantitative estimation of aqueous ferrous nitrate in solution.

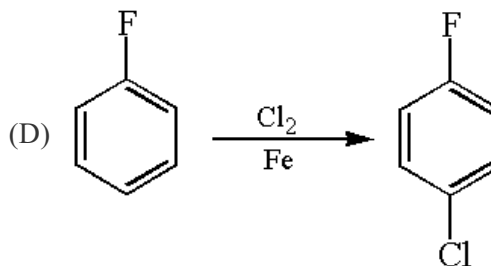
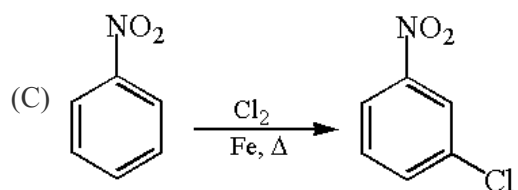
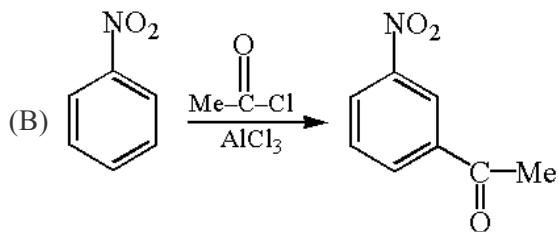
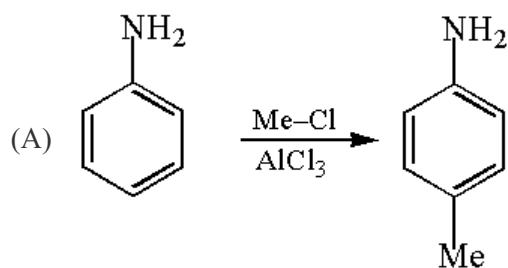


Which of the following option is correct regarding α and β ?



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11. Identify reactions which are not feasible.



12. For a first order reaction :

- (A) The degree of dissociation is equal to $(1 - e^{-kt})$
- (B) A plot of reciprocal concentration of the reactant vs time gives a straight line
- (C) The time taken for the completion of 75% reaction is thrice the $t_{1/2}$ of the reaction
- (D) The pre-exponential factor in the Arrhenius equation has the dimension of time, T^{-1}

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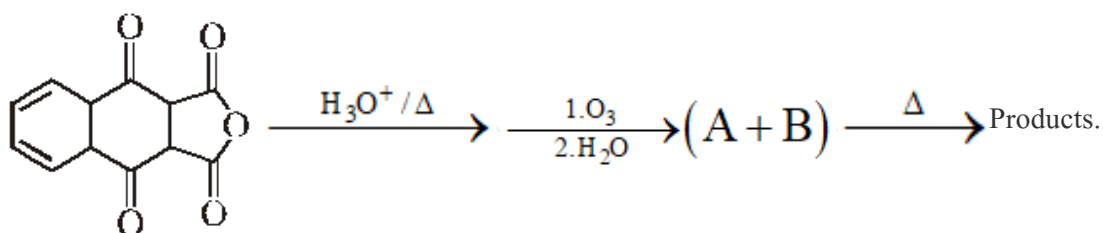
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Number of isomers possible for the complex compound $[\text{Co}(\text{en})_2\text{Cl}(\text{NO}_2)]\text{Cl}$ are _____.
2. Mole of CO_2 released if 1 mole reactant reacts in the following sequence.



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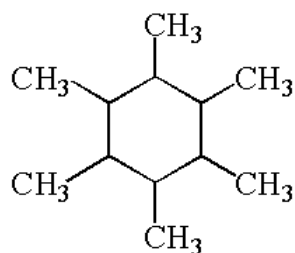
3. Minimum volume of 0.2 M $\text{KMnO}_4(V_1)$ and 1 M $\text{K}_2\text{Cr}_2\text{O}_7(V_2)$ required to completely oxidise same volume of known concentration of FeSO_4 separately would give V_1/V_2 as _____.
4. One of the Mica silicate is formed when one quarter of the Si^{IV} in talc are replaced by Al^{III} and resulting charge is balanced by K^{I} .

If formula of talc is $\text{Mg}_3(\text{OH})_2(\text{Si}_4\text{O}_{10})$.

Find sum of K^{I} , Mg^{II} & Si^{IV} atoms in formula of Mica.

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5. How many optically inactive geometrical isomers are possible ?



6. 620 g ethylene glycol is added to 4 kg water in the radiation of a car. The amount (in gm) of ice will separate out at -6°C is $x \times 100$ ($K_f = 1.86 \text{ K kg mol}^{-1}$.) Find x

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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- [illegible]

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3. The greatest value of the term independent of x in the expansion $\left(x \sin \alpha + \frac{\cos \alpha}{x}\right)^{2n}$ $\alpha \in \mathbb{R}$ is equal to
- (A) $\frac{{}^{2n}C_{n+1}}{2^{n+1}}$ (B) $\frac{{}^{2n}C_{n-1}}{2^{n-1}}$ (C) $\frac{{}^{2n}C_n}{2^n}$ (D) $\frac{{}^{2n}C_n}{2^{2n}}$
4. If the points $\left(\frac{a^3}{a-1}, \frac{a^2-3}{a-1}\right)$, $\left(\frac{b^3}{b-1}, \frac{b^2-3}{b-1}\right)$ and $\left(\frac{c^3}{c-1}, \frac{c^2-3}{c-1}\right)$ are collinear for distinct a , b and c , the value of $abc - (ab + bc + ca) + 3(a + b + c)$ is
- (A) 0 (B) 1 (C) -1 (D) None of these

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SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. $f(x) = \begin{cases} x + \frac{1}{2}, & x < 0 \\ 2x + \frac{1}{3}, & x \geq 0 \end{cases}$, identify the correct statement(s) ([] denotes greatest integer function)

(A) $\lim_{x \rightarrow 0^+} [f(x)] = 0$

(B) $\lim_{x \rightarrow 0} f(x)$ does not exist

(C) $\left[\lim_{x \rightarrow 0} f(x) \right]$ exist

(D) $\lim_{x \rightarrow 0} \frac{[f(x)]}{x}$ does not exist

6. If the equation $z^4 + a_1 z^3 + a_2 z^2 + a_3 z + a_4 = 0$, (where a_1, a_2, a_3, a_4 are non-zero real coefficients), has a purely imaginary root then the expression $\frac{a_3}{a_1 a_2} + \frac{a_1 a_4}{a_2 a_3}$ is less than or equal to

(A) $\frac{1}{2}$

(B) 2

(C) 3

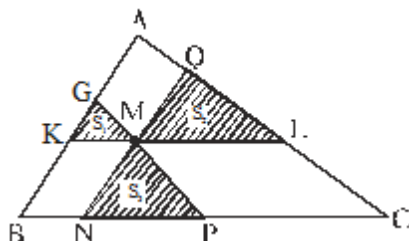
(D) $\frac{1}{3}$

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7. If the straight line $3x + 4y + 5 = 0$ touches a parabola at $(1, -2)$ with focus at $(4, 2)$, then :
- (A) its directrix passes through the point $(2, -9)$
 (B) there will be a unique parabola of such kind.
 (C) no such parabola exists
 (D) the length of latus rectum of such all parabolae will be 20.
8. If the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ intersects the line $3\beta^2x + 3(1 - 2\alpha)y + z = 3 = -\frac{1}{2}\{6\alpha^2x + 3(1 - 2\beta)y + 2z\}$, then point $(\alpha, \beta, 1)$ lie on the plane
- (A) $2x - y + z = 4$ (B) $x + y - z = 2$
 (C) $x - 2y = 0$ (D) $2x - y = 0$
9. The value of the limit $\lim_{n \rightarrow \infty} \frac{\int_0^{n\pi} \sin^4 x dx}{n\pi}$ is lesser than or equal to : ($n \in \mathbb{I}$)
- (A) $\frac{3}{16}$ (B) $\frac{3}{8}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

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10. Three straight lines are drawn through a point M, lying in the interior of triangle ABC, parallel to its sides. The areas of the resulting three triangles (see Fig.) are S_1 , S_2 and S_3 . The area of triangle ABC will not be equal to



- (A) $S_1 + S_2 + S_3$ (B) $(\sqrt{S_1} + \sqrt{S_2} + \sqrt{S_3})^2$
 (C) $\frac{(S_1 + S_2 + S_3)^{\frac{3}{2}}}{S_1 + S_2 + S_3}$ (D) $\sqrt{S_1 S_2} + \sqrt{S_2 S_1} + \sqrt{S_2 S_3}$
11. Let $f(x)$ be polynomial function satisfying $0 < xf(y) < yf(x) \forall x, y$ such that $0 < x < y \leq 1$ and $f(0) = 0$ then :

- (A) $f'(x) < f(1)$ (B) $f(1) < 2 \int_0^1 f(x) dx$
 (C) $3f\left(\frac{1}{3}\right) > 2f\left(\frac{1}{2}\right)$ (D) $6f\left(\frac{1}{6}\right) < 5f\left(\frac{1}{5}\right)$
12. Let M be a 2×2 symmetric matrix with integer entries. Then M is invertible if
- (A) the first column of M is the transpose of the second row of M
 (B) the second row of M is the transpose of the first column of M
 (C) M is a diagonal matrix with nonzero entries in the main diagonal
 (D) the product of entries in the main diagonal of M is not the square of an integer

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SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. The sum of how many terms of the sequence $\sum_{r=0}^n \frac{1}{(\sqrt{4r+1} + \sqrt{4r+5})(\sqrt[4]{4r+1} + \sqrt[4]{4r+5})}$ will be 1.5 ?
2. A curve $y^2 = f(x)$ satisfies the differential equation $2y \frac{dy}{dx} \sin x = \cos x (\sin x - y^2)$ and passes through the point $(\frac{\pi}{2}, 2)$. Evaluate $f(\frac{\pi}{6})$
3. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid point of the line segment joining the centres of C_1 and C_2 and C be a circle touching C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C_1 , then the radius of the circle C is

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4. A group of 6 boys and 6 girls is randomly divided into two equal groups. The probability that each group contains 3 boys and 3 girls is p . Find the value of p^{-1} .
5. In a ΔABC , r_1 , r_2 and r_3 denote the exradii opposite to vertices A, B and C respectively. If $r_1 = 2r_2 = 3r_3$ then evaluate : $(\sin B + \sin C)(\cos B + \cos C)$
6. The volume of a tetrahedron OABC formed by the vectors $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ & $\overrightarrow{OC} = \vec{c}$ where $|\vec{a}| = 3$, $|\vec{b}| = 4$, $|\vec{c}| = 6$, angle between the vector - pairs $\{\vec{a}, \vec{b}\}$; $\{\vec{b}, \vec{c}\}$ and $\{\vec{c}, \vec{a}\}$ be $\frac{\pi}{3}$, $\cos^{-1} \frac{1}{4}$ and $\cos^{-1} \left(\frac{-1}{18} \right)$ respectively is V. If the value of V^2 can be written as $\frac{pq}{r^2}$ where p, q, r are prime numbers then evaluate $\frac{p+q}{r}$.

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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